Electric Vehicle Management System Using Artificial Intelligence and Machine Learning

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***Abstract*—Electric Vehicles (EVs) are hampered by the issues of battery degradation, operational anomalies, range estimation uncertainty, charging time and access to quality vehicle information. Current EV use frequently involves independent use of functions like State of Charge (SOC) monitoring, location of charging stations, and simple range estimation, without integrated predictive and personalized decision support. This research suggests the development of a four-part interconnected mobile decision-support system (MDSS) based on Artificial Intelligence (AI). The first one identifies operational anomalies according to vehicle parameters like battery voltage, current, temperature, motor state, speed, acceleration, charging status and power consumption. The second will forecast battery State of Health (SOH) degradation for 3, 6 and 12-month time horizons and offer customized battery protection suggestions. The third enables intelligent EV mobility, with features such as range prediction, charging-station filtering, queue forecasting, route optimization, charging-time estimation and cost analysis. The fourth combines Retrieval-Augmented Generation (RAG), Large Language Models (LLMs) and voice/text interaction to enable context-aware EV guidance. The system's proposed integration of diagnostic, predictive, optimization, and conversational intelligence in a single mobile application suggests it could enhance EV safety, battery management, charging efficiency, and user decision-making.**

***Index Terms*—Electric Vehicles, Artificial Intelligence, Battery State of Health, Anomaly Detection, Charging and Range Optimization, Retrieval-Augmented Generation.**

I. Introduction

Sustainable transportation has boosted the use of electric vehicles (EVs). As of 2025, electric cars make up just over one-quarter of the total global sales of new cars [1]. The growth has also led to the growing need for intelligent technologies to help support EV users in managing battery condition, vehicle operation, driving range, charging, route planning, and access to technical information. For these areas battery management is especially critical, as it affects the range, performance, reliability, and life-cycle cost of the vehicle. Operating and environmental conditions cause battery degradation, which ultimately leads to loss of capacity and power [2].

Although the EV industry has made major advances in technology, users are still relying on individual apps or services to track their vehicles' State of Charge (SOC), estimate range, access charging station data and information, navigate, verify vehicle diagnostics, and determine maintenance needs. Simple SOC data does not necessarily imply changes in battery health over time, and a simple threshold-based warning does not necessarily reflect abnormalities associated with multiple vehicle parameters. Likewise, battery state, driving history, traffic, availability of charging stations, and other operational considerations may impact remaining range and charging decisions.

Some prior studies have examined battery-health estimation with machine learning, anomaly detection in operation, driving-range prediction, charging-station occupancy prediction, route optimization, and conversational assistance. These capabilities are typically researched or developed separately, however. As a result, an EV user might require multiple individual systems to learn about the condition of their vehicle, estimate battery life, plan their trip and charging schedule, and receive technical advice.

To overcome this problem, this research introduces the concept of an integrated mobile decision support system that consists of four independent components. AI-Based EV Operational Anomaly Detection, Adaptive Battery Health Prediction and Personalized Recommendation, Intelligent EV Charging Optimization and Range Prediction, AI-Based EV Virtual Assistant. The aim of this research is to build and test an AI- and machine learning-based EV decision support system that will

deliver diagnostic, predictive, optimization, and conversational support via one mobile app.

II. LITERATURE REVIEW

Capacity loss and power degradation are among the most critical issues in EV management, as they have a direct impact on the driving range, vehicle performance, and long-term operating cost for EVs, which is a great concern in the degradation of lithium-ion batteries [2]. Battery-health estimation and degradation prediction have thus been researched using data-driven means. Severson et al. have shown that features from the first few cycles of battery life are sufficient for the machine learning to predict the battery cycle life before significant capacity loss occurs [3]. Roman et al. proposed a charging voltage and current-based machine learning framework for battery SOH estimation [4]. Lu et al. also presented a deep learning and domain adaptation approach for the estimation of SOH in commercial batteries of various manufacturers [5]. While these studies prove the efficacy of battery-health estimation based on data, they are mainly directed toward real-time SOH estimation or battery cycle-life prediction without combining multi-horizon battery forecasts with user-specific behavioral recommendations.

Another crucial operational anomaly detection in intelligent EV management is significant. The abnormal vehicle conditions may be dependent on relationships between various electrical, thermal, charging, and dynamic parameters, and not just one parameter being above a threshold. Liu et al. proposed Isolation Forest as an efficient method for identifying rare abnormal observations [6], and Schölkopf et al. came up with an idea of learning the region corresponding to the normal observations and detecting the samples outside that region, One-Class Support Vector Machine (SVM) [7]. Autoencoder-based techniques have also been used in the field of battery abnormality detection, where Zhang et al. were able to detect abnormalities in real EV battery data even when there were only limited labeled battery fault samples available [8]. The three methods laid the groundwork for multidimensional analysis of EV operational data and generation of meaningful anomaly information.

The issues of driving range uncertainty and charging stations are still of concern to EV owners. Ma et al. studied the real-world EV range prediction and showed the effect of SOC, battery-health indicators, and driving patterns on range estimation [9]. Charging-station availability is another factor that impacts EV travel. Ma and Faye suggested a hybrid method based on LSTM to predict the occupancy of charging stations, leveraging the charging-state sequence for the past and the temporal information [10]. Houalef et al. [11] have suggested personalized, energy-aware EV route planning that takes into account both energy consumption and driving behavior, environment, and charging stops, rather than geographical distance only. While these studies enhance individual elements of EV mobility, the challenge of range prediction, queue prediction, route optimization, charging-cost estimation, and charger selection are typically approached as isolated functions.

Another chance to enhance access to technical information about EVs is conversational AI. Retrieval-Augmented Generation (RAG) is a process that involves the retrieval of external information and subsequent generation of the response, which is grounded in the retrieved information and not just the pretrained knowledge of a language model [12]. The Dense Passage Retrieval mechanism is a powerful approach to retrieve semantically relevant information from large collections of documents [13]. These techniques can be used to help develop an EV-specific conversational assistant that can provide EV battery information, including battery care, charging, maintenance, and efficient driving. However, the current studies focus mainly on isolated problems such as battery-health estimation, anomaly detection, range and charging optimization, and conversational support. IntelligentEV does that by creating a consistent mobile architecture, which brings these functions together in one decision support workflow.

III. METHODOLOGY

The proposed IntelligentEV system integrates four machine learning and artificial intelligence (AI) modules within one mobility decision support system. Depending on the data required, different data sources are used.Vehicle operational parameters are used for anomaly detection, historical battery data is used for SOH prediction. Behavioral information is used for personalized recommendations, mobility and charging information supports range and charging optimization and an EV-specific knowledge base supports conversational assistance.

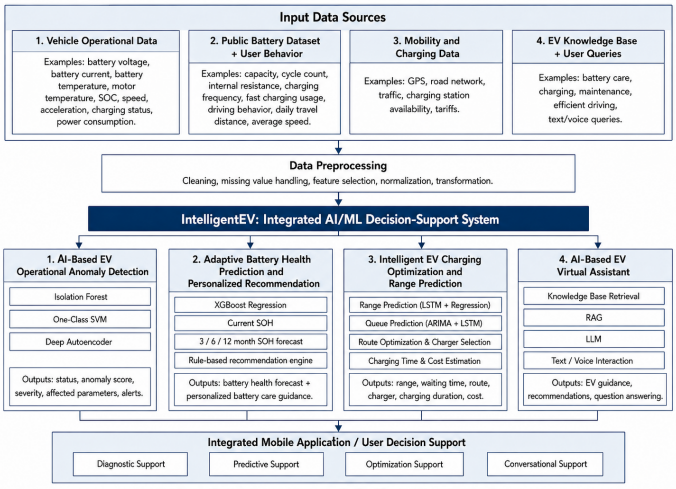


Fig. 1. Overall diagram of the proposed Intelligent EV system.

A. AI-Based EV Operational Anomaly Detection

1) Input and Preprocessing: The operational anomaly-detection module compares various EV parameters in combination to detect abnormal electrical, thermal, motor, charging, and power-consumption conditions. The variables being fed in are the battery voltage, battery current, battery temperature,

motor temperature, state of charge (SOC), vehicle speed, acceleration, charging state, and power consumption. By studying these variables together, the system can detect abnormal operating conditions that may not be apparent if the variables are considered individually.

The operational data is cleaned and converted to model-ready data before model analysis. Observations that were missing and/or inconsistent are dealt with, features are selected, and numbers are normalized if necessary to ensure consistent model inputs.

2) Anomaly Detection: Three methods of anomaly detection are explored. Isolation Forest, One-Class Support Vector Machine, and Deep Autoencoder. Isolation Forest uses random partitioning to find outlier observations and can be used to find rare abnormal operating records. One-Class SVM trains a boundary that describes normal EV operation and classifies points falling outside the boundary as possible anomalies. The deep autoencoder is trained to learn "normal" operating conditions and then classifies observations that are not reconstructed well as "abnormal."

3) Diagnostic Output: The model output is interpreted to give the user diagnostic information, not just a binary classification. The component generates vehicle status, anomaly score, anomaly category, severity level, affected parameters, diagnostic alerts, and recommended actions. Severity ranges from LOW, MEDIUM, and HIGH to CRITICAL, and anomaly categories include thermal, electrical, motor, charging, power-consumption, and unknown.

B. Adaptive Battery Health Prediction and Personalized Recommendation

1)Dataset and Preprocessing:

The battery-health component is comprised of battery degradation prediction and personalized recommendation generation. The battery degradation observations are obtained from historical data in a public battery data set, which are NCM (Nickel Cobalt Manganese) batteries. Relevant battery features are selected, and invalid, missing, or duplicate records are dealt with prior to model development. Numerical attributes are transformed into appropriate machine learning formats and the resulting data is split into training and test parts.

Where battery capacity information is available, the State of Health can be represented as

$$SOH(\%) = \frac{C_{\text{current}}}{C_{\text{rated}}} \times 100 \tag{1}$$

where C_{current} represents the current usable battery capacity and C_{rated} represents the original rated battery capacity.

2)XGBoost-Based Battery Degradation Prediction:

Nonlinear relationships between battery operating characteristics and SOH are modeled using XGBoost regression. XGBoost regression is used for modeling nonlinear relationships between battery operating characteristics and SOH. The algorithm builds an ensemble of decision trees one after the other, where each tree corrects errors made by the previous tree. The trained model is applied to predict the battery-health trend and then to aid in the SOH prediction at three different time horizons: 3 months, 6 months and 12 months.

The coefficient of determination (R^2), mean absolute error (MAE), and root mean square error (RMSE) are used to judge the model performance. The unseen testing dataset is used to assess the generalization of the model during training.

3)Personalized Recommendation Generation:

User charging and driving behavior is not part of the XGBoost training process since the battery dataset used to build the model does not contain them. Information input to the recommendation engine includes charging frequency, fast-charging utilization, charging behavior, daily driving distance, driving behavior, and average speed range.

These inputs are transformed into individual recommendations, thanks to predefined behavioral rules. For instance, if a driver is seen making unnecessary rapid speed increases frequently, they will be encouraged to slow down during normal driving and will see fewer unnecessary rapid speed increases. If a driver's driving patterns include aggressive braking more often than normal, they will receive warnings to brake more gently during normal driving and will see fewer aggressive braking episodes. Battery-health information can also be used to help prioritize recommendations. The output of the final stage, then, includes battery-health details, future battery-service health predictions, and user-specific battery-protection instructions.

C. Intelligent EV Charging Optimization and Range Prediction

1) Smart Driving Range Prediction: The range-prediction module calculates the distance that can be traveled on the battery's remaining charge under dynamic driving situations. The input parameters comprise SOC, speed over time, ambient temperature, battery temperature, elevation, auxiliary power consumption, etc., and other context-specific vehicle information. Noise in the sensors is minimized, and missing observations in the time series are filled in before model computation. This hybrid LSTM and regression method consider the temporal and tabular relationships between the selected variables and determine the estimated range of driving.

2) Smart Queue and Availability Prediction: The queue-prediction module throws an estimate of how busy the charging station is and how long it will take to charge. Input consists of real-time charger-status data, past charging-session information, plug availability, time-of-day charging trends, day-of-week occupancy, charging-rate type, and historical station turnaround time. To capture the recurring occupancy patterns, along with their short-term fluctuations in the number of chargers used, an ARIMA-LSTM forecasting approach is employed.

3) Route Optimization and Charger Selection: A weighted spatial network is used to represent road network information, traffic information, and charging-station locations. Route-selection factors are the travel time, expected energy consumption, charge station wait time, charger capacity, and the user's preferences. Multi-criteria station selection is integrated

with A* and Dijkstra-based path finding to find the suitable routes and charging stops based on the vehicle range and user requirements.

4) Charging Time and Cost Estimation: The charging-time and cost module takes into account initial SOC, target SOC, battery capacity, charger power, electricity tariffs, charging-station service fees, and non-linear charging behavior of the batteries. Estimation of charging duration is based on a model of integration using CC-CV. Charging cost is then calculated based on the estimated time and the tariff information to arrive at a charging cost. As a result, charging-time and cost data is made available via the application dashboard, which can be utilized in charging-planning decisions.

D. AI-Based EV Virtual Assistant

The AI-based EV virtual assistant will be a mix of an EV-specific knowledge base, retrieval-augmented generation (RAG), a large language model, and voice interaction. The knowledge base includes a body of structured knowledge on battery management, charging methods, battery health, efficient driving and maintenance, and frequent EV-related questions.

Upon the user's query, the language model is provided with relevant information from the knowledge base as well as the user query itself. With the RAG architecture, generated responses will be able to leverage EV-specific contextual information, rather than just the pretrained knowledge of the LLM, which will help make the generated guidance more relevant.

The assistant can interact with you via text and voice. The spoken query can be converted to text, sent through the RAG/LLM pipeline, and returned in the conversational interface as text or voice output. If available, the user's needs and the vehicle context may also be taken into account to give personal guidance on charging, battery maintenance, efficient driving, and maintenance.

IV. RESULTS AND DISCUSSION

A. AI-Based EV Operational Anomaly Detection

Various vehicle operating situations were used to assess the anomaly-detection component. The representative results obtained under normal and abnormal operating conditions are summarized in Table I.

TABLE I
OPERATIONAL ANOMALY DETECTION RESULTS

Operating Condition	Anomaly Score	Classification	Severity
Normal operation	0.08	NORMAL	None
Charging-system anomaly	0.96	ABNORMAL	CRITICAL
Motor/thermal anomaly	0.91	ABNORMAL	HIGH

As shown in Table I, the system generated a low anomaly score of 0.08 under normal operating conditions and classified the vehicle as NORMAL with no severity level. In contrast, the charging-system anomaly produced an anomaly score of 0.96 and was classified as CRITICAL. The motor and thermal irregularity produced an anomaly score of 0.91 and was classified as HIGH severity.

The results demonstrate the importance of multi-parameter anomaly analysis. Instead of producing only a binary normal or abnormal result, the component provides diagnostic information that helps the user understand the type and severity of the detected abnormality. This makes the output more suitable for early fault investigation and immediate vehicle-management decisions.

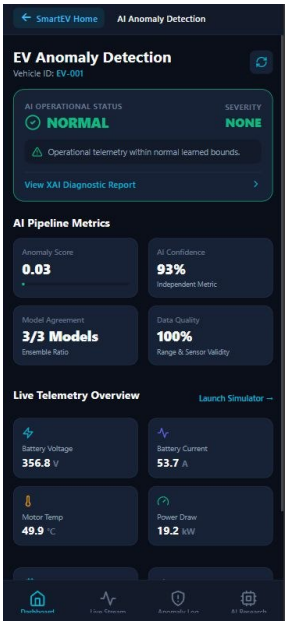


Fig. 2. Mobile application interface for AI-Based EV Operational Anomaly Detection.

Figure 2 shows the mobile application interface used to present the anomaly-detection results. The interface displays the detected vehicle condition and related diagnostic information, allowing the user to interpret the anomaly result through the mobile application.

B. Adaptive Battery Health Prediction and Personalized Recommendation

The battery-health prediction component was evaluated for three forecasting horizons: 3 months, 6 months, and 12 months. Random Forest and XGBoost regression models were independently trained and evaluated for each forecasting horizon. Model performance was assessed using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2). The independent test-set results are summarized in Table II.

TABLE II
BATTERY-HEALTH MODEL COMPARISON

Horizon	Model	MAE	RMSE	R^2
3 Months	Random Forest	0.017522	0.023360	0.999720
3 Months	XGBoost	0.019656	0.026104	0.999651
6 Months	Random Forest	0.028245	0.039355	0.998999
6 Months	XGBoost	0.030581	0.043160	0.998797
12 Months	Random Forest	0.024652	0.029916	0.999170
12 Months	XGBoost	0.030834	0.036723	0.998749

As shown in Table II, both Random Forest and XGBoost achieved high predictive performance across all three forecasting horizons. However, Random Forest consistently produced lower MAE and RMSE values and slightly higher R^2 values than XGBoost on the independent test datasets.

For the 3-month prediction horizon, Random Forest achieved an MAE of 0.017522, an RMSE of 0.023360, and an R^2 of 0.999720, compared with an MAE of 0.019656, an RMSE of 0.026104, and an R^2 of 0.999651 for XGBoost.

For the 6-month prediction horizon, Random Forest achieved an MAE of 0.028245, an RMSE of 0.039355, and an R^2 of 0.998999, whereas XGBoost achieved an MAE of 0.030581, an RMSE of 0.043160, and an R^2 of 0.998797.

For the 12-month forecasting horizon, Random Forest achieved an MAE of 0.024652, an RMSE of 0.029916, and an R^2 of 0.999170. XGBoost obtained an MAE of 0.030834, an RMSE of 0.036723, and an R^2 of 0.998749.

Based on the independent test-set comparison, Random Forest was selected as the final battery-health prediction model because it provided the lowest prediction errors and the highest R^2 values across the 3-, 6-, and 12-month forecasting horizons.

The personalized recommendation engine was evaluated separately from the machine-learning models because user charging and driving behavior variables were not included in the battery dataset used for model training. Different charging and driving patterns were assessed using predefined behavioral rules. Frequent fast charging generated recommendations to reduce unnecessary DC fast-charging sessions, while aggressive driving behavior resulted in recommendations for smoother acceleration and braking.

The final component therefore combines Random Forest-based SOH forecasting for 3-, 6-, and 12-month horizons with personalized battery-care recommendations generated according to the user's charging and driving behavior.

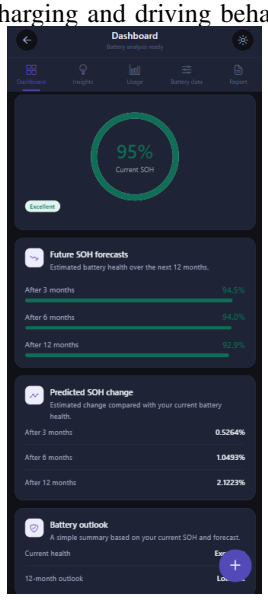


Fig. 3. Mobile application interface for Adaptive Battery Health Prediction and Personalized Recommendation.

Figure 3 presents the implemented mobile application interface of the battery-health component. The interface displays battery-health information and the 3-, 6-, and 12-month SOH forecasting outputs together with personalized battery-care recommendations for the user.

C. Intelligent EV Charging Optimization and Range Prediction

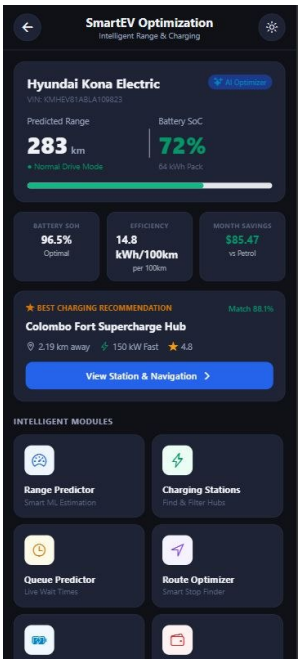


Fig. 4. Mobile application interface for intelligent EV charging optimization and range prediction.

Figure 4 presents the mobile application dashboard developed for this component. The interface displays key vehicle information together with functions for range prediction, charging-station selection, queue prediction, route optimization, and charging support. This integrated presentation allows users to access the main charging and mobility functions through a single mobile interface.

The charging and mobility component was evaluated based on driving-range prediction, charging-station queue prediction, route optimization, and charging-time and cost estimation. The main evaluation results are summarized in Table III.

TABLE III
CHARGING AND MOBILITY COMPONENT RESULTS

Function	Evaluation Metric	Result
Driving-range prediction	RMSE	4.12 km
Driving-range prediction	MAE	3.05 km
Queue prediction	MAE	2.4 min
Route optimization	Performance score	94.2%
Charging-time/cost estimation	Performance score	93.6%
Charging-time estimation	Variation margin	$\pm$3.5 min

As shown in Table III, the smart driving-range prediction module achieved an RMSE of 4.12 km and an MAE of 3.05 km on the validation dataset. The inclusion of auxiliary power consumption and elevation information helped the model account for variations in vehicle energy consumption.

The ARIMA-LSTM queue-prediction module achieved an MAE of 2.4 minutes and was able to capture recurring charging-station occupancy patterns and short-term variations. The route-optimization approach considered traffic conditions, detour distance, predicted waiting time, and charger capacity, and reported a performance score of 94.2%.

The charging-time and cost-estimation module, based on the CC-CV charging model, reported a performance score of 93.6%, with an estimated charging-time variation margin of ± 3.5 minutes. Overall, these results indicate that the component can integrate range estimation, charging-station availability, route planning, and charging information to support EV travel and charging decisions.

D. AI-Based EV Virtual Assistant

The virtual-assistant prototype successfully combined RAG (LLM) and voice interaction. The system pulled relevant information from the EV-specific knowledge bank and utilized it to formulate understandable responses concerning battery health, charging, maintenance, and efficient driving. There was interaction in the form of text and speech, with the user able to get a message without having to get through complicated application interfaces.

The prototype was easy to use and was perceived to have practical aspects, voice interaction and response speed were perceived to be beneficial for users, as indicated in user feedback. The quality of the responses, however, is contingent on the knowledge base of the EV being complete and up-to-date. Environmental noise, unclear speech, pronunciation, and language differences also can impact voice-recognition performance.

V. CONCLUSION AND FUTURE WORK

This research introduced IntelligentEV, a mobile intelligent decision support system based on AI and machine learning that is integrated with the EV management system. Solutions that focus on battery monitoring, operational fault detection, range estimation, charging support, and technical assistance are distinctly different from IntelligentEV, which is an all-in-one solution with four interconnected components: AI-Based EV Operational Anomaly Detection, Adaptive Battery Health Prediction and Personalized Recommendation, Intelligent EV Charging Optimization and Range Prediction, and an AI-Based EV Virtual Assistant.

The evaluation proved the capabilities could be integrated into a single system. Operational anomaly-detection functionality detected the normal and abnormal vehicle conditions and provided anomaly scores, severity levels, affected parameters, and actionable diagnostic guidance. The test R^2 was 0.9970, MAE was 0.81, and RMSE was 1.04 SOH percentage points for the battery-health model on the used dataset. The personalized recommendation engine produced charging and driving directions as per the behavior of the user. The charging and mobility part assisted in charging-range prediction, charging-station queue prediction, route selection, charger selection, and charging-time and cost estimation. The virtual assistant based on RAG and LLM was also used to deliver EV information via text and voice, in context.

Limited by datasets, prototype environments, predefined recommendation rules, and available knowledge sources in this study, the reported performance is limited. Future studies should therefore test the system with larger and more representative real-world EV datasets comprising various vehicle models, battery chemistries, environments, and driving scenarios. The battery-health model needs to be cross-validated with the independent longitudinal battery data to evaluate the model's performance on the 3, 6 and 12-month SOH forecasting under the realistic operation environment.

Real-time vehicle telemetry, the charging station API, traffic and weather data, electricity tariffs, and charging network data can be integrated into further development. Adaptive learning could be used to learn from user behavior and battery health trends to replace or augment the predefined rules for recommendations. The virtual assistant in the future may also be able to enable multilingual text and voice interaction, enhance speech recognition, broaden the EV knowledge base, and strengthen the integration of the predictive outputs of the other system components. Furthermore, long-term field testing is needed with EV users to check reliability, usability, response accuracy, computational efficiency, and the effectiveness of the recommendations.

REFERENCES

[1] International Energy Agency, *Global EV Outlook 2026*. Paris, France: IEA, 2026.

[2] J. S. Edge, S. O'Kane, R. Prosser, N. D. Kirkaldy, A. N. Patel, A. Hales, A. Ghosh, W. Ai, J. Chen, J. Yang, S. Li, M.-C. Pang, L. Bravo Diaz, A. Tomaszewska, M. W. Marzook, K. N. Radhakrishnan, H. Wang, Y. Patel, B. Wu, and G. J. Offer, "Lithium ion battery degradation: What you need to know," *Physical Chemistry Chemical Physics*, vol. 23, pp. 8200–8221, 2021, doi: 10.1039/D1CP00359C.

[3] K. A. Severson, P. M. Attia, N. Jin, N. Perkins, B. Jiang, Z. Yang, M. H. Chen, M. Aykol, P. K. Herring, D. Fraggedakis, M. Z. Bazant, S. J. Harris, W. C. Chueh, and R. D. Braatz, "Data-driven prediction of battery cycle life before capacity degradation," *Nature Energy*, vol. 4, pp. 383–391, 2019, doi: 10.1038/s41560-019-0356-8.

[4] D. Roman, S. Saxena, V. Robu, M. Pecht, and D. Flynn, "Machine learning pipeline for battery state-of-health estimation," *Nature Machine Intelligence*, vol. 3, pp. 447–456, 2021, doi: 10.1038/s42256-021-00312-3.

[5] J. Lu, R. Xiong, J. Tian, C. Wang, and F. Sun, "Deep learning to estimate lithium-ion battery state of health without additional degradation experiments," *Nature Communications*, vol. 14, Art. no. 2760, 2023, doi: 10.1038/s41467-023-38458-w.

[6] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation Forest," in *Proc. 8th IEEE Int. Conf. Data Mining (ICDM)*, Pisa, Italy, 2008, pp. 413–422, doi: 10.1109/ICDM.2008.17.

[7] B. Schölkopf, J. C. Platt, J. Shawe-Taylor, A. J. Smola, and R. C. Williamson, "Estimating the support of a high-dimensional distribution," *Neural Computation*, vol. 13, no. 7, pp. 1443–1471, 2001, doi: 10.1162/089976601750264965.

[8] X. Zhang, P. Liu, N. Lin, Z. Zhang, and Z. Wang, "A novel battery abnormality detection method using interpretable Autoencoder," *Applied Energy*, vol. 330, Art. no. 120312, 2023, doi: 10.1016/j.apenergy.2022.120312.

[9] X. Ma, J. Li, H. Cui, L. Cheng, Y. Ji, and J. Wang, "Electric vehicle range prediction considering real-time driving factors and battery capacity index," *Transportation Research Part D: Transport and Environment*, vol. 144, Art. no. 104795, 2025, doi: 10.1016/j.trd.2025.104795.

[10] T.-Y. Ma and S. Faye, "Multistep electric vehicle charging station occupancy prediction using hybrid LSTM neural networks," *Energy*, vol. 244, Art. no. 123217, 2022, doi: 10.1016/j.energy.2022.123217.

[11] A.-R. Houalef, F. Delavernhe, S.-M. Senouci, and E.-H. Aglzim, "Data-driven, personalized route planning for connected electric vehicles: Optimizing time, energy, and charging stops," *Applied Energy*, vol. 402, Art. no. 126887, 2025, doi: 10.1016/j.apenergy.2025.126887.

[12] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems*, vol. 33, pp. 9459–9474, 2020.

[13] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, "Dense Passage Retrieval for Open-Domain Question Answering," in *Proc. 2020 Conf. Empirical Methods in Natural Language Processing (EMNLP)*, pp. 6769–6781, 2020, doi: 10.18653/v1/2020.emnlp-main.550.